



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

COMPUTER SCIENCE

0478/11

Paper 1

October/November 2019

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	Printer Statement <pre> graph LR IP[Inkjet printer] --- C[Can print in colour] IP --- L[Creates output line by line using a print head] LP[Laser printer] --- C LP --- D[Uses a charged drum to create the printed item] LP --- T[Uses powdered toner] </pre> One mark for correct lines from inkjet One mark for correct lines from laser	2
1(b)	Laser	1
1(c)	Two from: - Design is created on the computer / software / CAD - Material is loaded to cutter - Different types of material can be used - Uses lasers to cut material that use infra-red ... that produces extreme heat ... that is focussed using a special lens - Can work on both the x,y and z axis	2

Question	Answer	Marks								
2(a)	One mark for each correct denary value <table><tr><th>Binary</th><th>Denary</th></tr><tr><td>0001001110</td><td>78</td></tr><tr><td>0110110111</td><td>439</td></tr><tr><td>1000000001</td><td>513</td></tr></table>	Binary	Denary	0001001110	78	0110110111	439	1000000001	513	3
Binary	Denary									
0001001110	78									
0110110111	439									
1000000001	513									
2(b)	Two from: Uses fewer characters // shorter Easier to read / write / understand Less likely to make mistakes // less error prone Easier to debug	2								
2(c)	One mark for each correct hexadecimal value in correct order 2 B 5	3								

Question	Answer	Marks
3(a)	Three from: Malicious software // type of malware Tracks / records keypresses // keylogger Sends data to third party Collected data is analysed to obtain data	3
3(b)	One mark for identified method, one mark for how it prevents spyware: Drop-down boxes // onscreen / virtual keyboard Means key logger cannot collect data Only requires part of the password Hacker doesn't get the full password Two-step verification // Two-factor authentication Extra data is sent to device making it more difficult for hacker to obtain it Data has to be entered into the same system // if attempted from a remote location, it will not be accepted Use a biometric device The person's biological data (e.g. their fingerprint) is also required	2
3(c)	Four from: Monitors traffic coming into and out of the computer system Checks that the traffic meets any criteria / rules set Blocks any traffic that does not meet the criteria / rules set Allows a set blacklist / whitelist // can block certain IP addresses Can close certain ports	4
3(d)	Two from: Passwords // biometrics Levels of access Proxy servers Physical security methods – e.g. PC's in locked rooms, etc.	2

Question	Answer	Marks
4(a)	One from: Touch screen Keyboard Microphone Mouse	1
4(b)	One from: Headphones Speakers Printer Light / LED	1
4(c)	One from: HDD SSD USB drive	1
4(d)	Four from: QR code is scanned using a <u>camera</u> on a mobile device and read / decoded using an application / software Illuminator shone on code Squares reflect light differently Corners of code are used for orientation Opens document with information // Directs to website with information QR code can be saved for future reference	4

Question	Answer	Marks
5(a)	Data is sent down a single wire one bit at a time Data is sent in both directions but only one direction at a time	4
5(b)	One mark for correct byte (Byte) 2 // 01010100 Three from: Added up / counted the 1s / 0s Even parity used // 3 bytes are even Byte 2 uses odd parity // 1 byte is odd	4

Question	Answer	Marks
6(a)(i)	One from: Code will run without the need of an interpreter (Object) Code is platform independent Source code not available / cannot be modified	1
6(a)(ii)	One from: Source code not available / cannot be modified Comments, etc. not visible Future changes will require code to be recompiled	1
6(b)(i)	One from: Protocol is HTTPS Padlock icon is locked Can view website certificate	1
6(b)(ii)	Five from: Browser / client sends request to webserver to request identification Web server sends its digital / security certificate Browser authenticates certificate if authentic connection, is established Any data sent is encrypted using public and private keys	5
6(c)	Four from: A type of software licence Free of charge Normally distributed without the source code Can legally share / copy Cannot legally modify code Cannot resell	4

Question	Answer	Marks
6(d)(i)	Two from: File size is reduced so it uses less storage space ... so faster transmission ... so quicker to download	2
6(d)(ii)	Lossless	1

Question	Answer	Marks															
7(a)	<table border="1"> <thead> <tr> <th>Input A</th><th>Input B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </tbody> </table>	Input A	Input B	Output	0	0	0	0	1	1	1	0	1	1	1	1	1
Input A	Input B	Output															
0	0	0															
0	1	1															
1	0	1															
1	1	1															
7(b)	Exclusive OR / XOR / EOR	1															
7(c)	One mark for each correct logic gate with correct inputs	5															
7(d)	Two from: - Can work continuously - Avoids human error - It could be a dangerous environment and will avoid human risk - Detect errors instantly - Maintain consistent and correct conditions	2															

Question	Answer	Marks
8	Six from: PC holds address of the instruction The address held in PC is sent to MAR using address bus MAR goes to location in memory where instruction is stored Instruction sent to MDR using data bus Instruction sent to CIR Control unit sends signals to manage the process using the control bus	6
9(a)	Two from: Layout of the webpage e.g. where a paragraph is placed Defined using tags	2
9(b)	One mark for each correct term in the correct order: browser IP address web server HTML	4